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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROGRAMMING PROJECT PRESENTATION

Project Title: Hospital Bed Allocation and Availability Monitoring System

Course: Java Programming Lab / Mini Project

Student Name & Roll No: [Yukesh . M &25BCS162]

Department: B.Sc.Computer Science

Mentor / Guide: [Name & Designation]

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INTRODUCTION

Hospitals need efficient systems to manage bed allocation and avoid overcrowding.

Manual tracking can lead to errors and delays.

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OBJECTIVE

Main goals of the project:

Track bed availability

Allocate beds to patients

Release beds

Improve hospital efficiency

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PROBLEM STATEMENT

Manual bed management causes delays, errors,and inefficency.

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SYSTEM REQUIREMENTS

Hardware Requirements:

- Processor: Intel i3 or higher
- RAM: 4 GB minimum
- Storage: 500 MB free space

Software Requirements:

- OS: Windows / Linux
- Compiler: JDK,Java console environment.

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SYSTEM DESIGN

Bed Class

Attributes:

Bed ID

Status (Available/Occupied)

Patient Name

Methods:

Allocate

Release

Display

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MODULE DESCRIPTION

- Input Module :
  - Number of beds
  - Bed ID selection
  - Patient name/detail
- Processing Module :
  - Checking bed availability
  - Allocating bed to a patient
- Output Module
  - Showing available and occupied beds
  - Displaying patient allocation details
- File Handling module:

  Saving bed and patient data

  Loading existing data when system starts

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ALGORITHM / FLOWCHART

1. Start
2. Input
3. Process
4. Display
5. Repeat
6. End

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IMPLEMENTATION

- Developed using java with oop and menu-driven program.

View bed availability

Allocate a bed to a patient

Release an occupied bed

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OUTPUT SCREENS

- Display Bed Status
- Allocation
- Discharge Status

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RESULTS & DISCUSSION

- System work efficiently and reduces manual errors.

The system successfully manages bed allocation, release, and availability with accurate results and user-friendly interaction.

The system helps reduce manual work and improves efficiency in managing hospital beds.

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CONCLUSION & FUTURE ENHANCEMENT

Conclusion:

This system improves hospital management by efficiently tracking and allocating beds, reducing workload and increasing accuracy.

Future Enhancements:

GUI Interface (Java Swing/JavaFX)

Database Integration (MySQL)

Online Monitoring System

Mobile application support

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REFERENCES / ACKNOWLEDGEMENT

References:

Java Programming – Herbert Schildt Java: The Complete Reference

Oracle Java Documentation

GeeksforGeeks – Java Concepts & Examples

W3Schools – Basic Java Syntax

Lecture notes and class material

THANK YOU

Questions & Discussions:

What is a Hospital Bed Allocation System?

Why is this system needed in hospitals?

Which programming language did you use and why?